

微积分(三重积分线面积分部分)综合训练题参考答案

1.(1)16,(2)16.

2. $9+\frac{15}{4}\ln 5$

3. $8\pi a^4$

4. $(\sqrt{2}+\frac{3}{2})\pi$

5. $2\pi \arctan \frac{H}{R}$

6. $(\sqrt{6}+2)\pi$

7. 2π

8.(1) $-\frac{\pi}{4}$ (2) $\frac{3\pi}{4}$ (3) $\frac{\pi}{2}$ (4) $\frac{\pi}{2}+\frac{1}{2}\arctan 3$

9. $1-\sqrt{2}(\lambda=-\frac{1}{2})$

10.(1) $-\frac{\pi}{4}a^3$ (2) $-\frac{\pi R^2}{2\sqrt{2}}$

11.(1)1.(2) $e-\frac{e^{16}}{16}$

12.(1) $x\sin \frac{1}{y}$ (2) $x\sin \frac{x}{y}$ (3) $\frac{x}{x+y}$ (4) $\frac{x-y}{(x+y)^2}$

(5) $x\sin(xy)$ (6) $\int_0^{xy} f(t)dt$ (7) $\frac{1}{2}\int_0^{x^2+y^2} f(t)dt$

13. $\frac{2}{3}$

14. $2\pi(e-e^{\sqrt{2}})$

15. -12π

16. 2π

17. $2(e^{2a}-1)\pi a^2$

18. $\frac{15}{16}(\pi+2)\pi$

19. 12π

$$20.(1)0.(2)\frac{256}{75}$$

$$21.(\frac{81}{4}+\frac{94}{5})\pi$$

$$22.40\pi$$

$$23.\frac{3\pi}{2}\ln 2$$

$$24.\frac{(\sqrt{2}-1)\pi}{6}$$

$$25.\frac{2}{3}\pi h^3 t + 2\pi h t f(t^2), \frac{1}{3}\pi h^3 + \pi h f(0)$$

$$26.(1)\text{提示: 令 } F(u) = \int_0^u f(t)dt, \text{ 则 } F(0)=0, F'(u)=f(u).....$$

$$(2)\text{提示: 原式} = \int_0^1 dx \int_0^x dy \int_0^y \frac{\sin z}{(1-z)^2} dz = \dots = \frac{1-\cos 1}{2}$$

$$(3) = \int_0^1 dy \int_y^1 dx \int_0^{xy} f(x, y, z) dz$$

$$= \int_0^1 dy \left[\int_0^{y^2} dz \int_y^1 f(x, y, z) dx + \int_{y^2}^y dz \int_{\frac{z}{y}}^1 f(x, y, z) dx \right]$$

$$(4) = \int_0^1 dx \left[\int_0^{x^2} dz \int_0^1 f(x, y, z) dy + \int_{x^2}^{1+x^2} dz \int_{\sqrt{z-x^2}}^1 f(x, y, z) dy \right]$$

$$27.(1)(0, 0, \frac{4}{5}h) \text{ (选择圆锥顶点为坐标原点)}$$

$$(2)(0, 0, \frac{R}{2})$$

$$(3)\frac{28}{15}\pi a^5$$

$$(4)(0, \frac{2k}{R})(k \text{ 为引力常数})$$